

Generation Interconnection System Impact Study Report

For Queue Project Q20-1486

Shady Oaks Wind Farm 100 kV

April 19, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1486. The Interconnection Customer proposes a 84.2 MW battery energy storage facility interconnecting at the Shady Oaks Wind Farm 100 kV substation in the Illinois area. The request was received on June 02, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 575 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$472,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

ACADIA to MONTOUR STA 500.0 kV Line: 102.4% loading (3,248 MVA) following the outage of the CORONET to WESTWOOD GENERATION LLC 500.0 kV Line; DFAX = 7.6%.

For the loss of PUEBLO XTRA to DAIRYMANS 230.0 kV #2, the PUEBLO XTRA to DAIRYMANS 230.0 kV Cir. 1 loads to 100.9% of its 885 MVA rating. Project DFAX: 7.9%.

For the loss of HILLTOP to TALENENERGY MARTINS CREEK LLC WILLIAMSPORT SUBSTATION 500.0 kV L, the HILLTOP SUBSTATION to STRAWBERRY BELT 345.0 kV loads to 100.3% of its 1,890 MVA rating. Project DFAX: 5.5%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MONTOUR STA to ACADIA 500.0 kV	Reconductor line	\$53,771,000	0% (\$277,000)
PUEBLO XTRA to DAIRYMANS 230.0 kV Cir. 1	Reconductor line	\$30,953,000	0% (\$143,000)
HILLTOP SUBSTATION to STRAWBERRY BELT 345.0 kV	Reconductor line	\$10,001,000	0% (\$52,000)
	Total allocated to project		\$472,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.